

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

**PRAGYA G8 – ACHIEVERS | MATHEMATICS
LOGARITHMS | LEVEL 4***JEE pattern – mixed difficulty, exam standard.*

1. If $\log_2 9 + \log_9 2 = k$, then the largest integer less than k is
(A) 3 (B) 1
(C) 4 (D) 2
2. If $x = \sqrt{\log_a b}$, $y = \sqrt{\log_b a}$ with $a > 0$, $b > 0$, $a, b \neq 1$, then $x^a - y^{-b} = 0$ implies
(A) a and b cannot be equal
(B) $a + b$ can be zero
(C) a and b must be equal
(D) $a + b$ must be zero
3. If a, b, c are positive reals with $a^2 + b^2 = c^2$, then $c + b \neq 1$ and $c - b \neq 1$, then $\log_{c+b} a + \log_{c-b} a$ equals
(A) $\log_{c+b} a \cdot \log_{c-b} a$ (B) 1
(C) $2 \log_{c+b} a \cdot \log_{c-b} a$ (D) 0
4. If $\frac{\log_2 a}{2x + 3y} = \frac{\log_2(b + 1)}{5y + 6z} = \frac{\log_2(c + 2)}{3z + 5x}$ (wherever defined) with $(a - 1)b(c + 1) \neq 0$ and $abc + 2ab + ac + 2a - 1 = 0$, then $(7x + 8y + 9z)$ equals
(A) 6 (B) 24
(C) 0 (D) 1
5. Which of the following is the greatest?
(A) $\log_{15} 36$ (B) $\log_4 6$
(C) $\log_3 5$ (D) $\log_{25} 36$
6. Which of the following is the smallest?
(A) $\log_5 500$ (B) $\log_4 192$
(C) $\log_6 1080$ (D) $\log_3 108$
7. Identify the correct order.
(A) $\log_3 6 < \log_3 8 < \log_2 5 < \log_2 6$ (B) $\log_2 5 < \log_2 6 < \log_3 6 < \log_3 8$
(C) $\log_3 8 < \log_3 6 < \log_2 6 < \log_2 5$ (D) $\log_2 6 < \log_3 6 < \log_3 8 < \log_2 5$

8. Which of the following is the largest?
- (A) $\log_4(\log_3(\log_2 5))$ (B) $\log_2(\log_4(\log_3 5))$
(C) $\log_2(\log_3(\log_4 5))$ (D) $\log_4(\log_3(\log_3 5))$
9. Let (x_0, y_0) solve the system $(5(x+1))^{\ln 5} = (2y)^{\ln 2}$ and $(x+1)^{\ln 2} = 5^{\ln y}$. Then x_0 is
- (A) $-\frac{2}{5}$ (B) $\frac{1}{5}$
(C) $\frac{4}{5}$ (D) $-\frac{4}{5}$
10. Let m be the number of positive integers whose logarithm to the base 3 has characteristic 4, and n the number of zeros after the decimal point and before the first significant digit of 2^{-100} . The remainder when m is divided by n is
- (A) 11 (B) 6
(C) 7 (D) 12
11. The number of solutions of the equation $\log_{11}(x^4 + 5) = \log_4(1 - x^6)$ (wherever defined) is
- (A) 0 (B) 4
(C) 6 (D) 1
12. The number of integers whose reciprocals have logarithm to the base 10 with characteristic $-n$ is
- (A) $9 \cdot 10^n$ (B) $9 \cdot 10^{n-1}$
(C) $9 \cdot 10^{-n}$ (D) $9 \cdot 10^{n+1}$
13. Let $f(x) = 4^{x+1.5} + 9^{x+0.5} - 10 \cdot 6^x$. The sum of the values of x satisfying $f(x) = 0$ is
- (A) $\log_{2/3}\left(\frac{3}{8}\right)$ (B) $\log_2 5$
(C) $\log_{3/4}\left(\frac{5}{9}\right)$ (D) $\log_7 11$
14. Let $g(x) = \log_{10}(2^x + 1) - \log_{10} 6 - x \log_{10} 5$. The value of x satisfying $g(x) = -x$ is
- (A) 3 (B) 2
(C) 0 (D) 1
15. Let $f(x) = 4^{\log_7 x} + 3^{\log_7 x}$. The number of solutions of $f(x) = x$ is
- (A) 2 (B) 1
(C) 3 (D) infinite
16. $\log_2 \left(\sqrt{2 - \sqrt{5}} + \sqrt{21 - 4\sqrt{5}} + \sqrt{14 - 6\sqrt{5}} \right) =$

(A) 4

(B) 3

(C) 1

(D) 2

17. The natural solution of $\frac{3x^4 + x^2 - 2x - 3}{3x^4 - x^2 + 2x + 3} = \frac{6x^4 + 2x^2 - 7x + 3}{6x^4 - 2x^2 + 7x - 3}$ is

(A) 3

(B) 1

(C) 2

(D) 4

18. The value of $3^{\log_{12} 60} \cdot 4^{\log_{12} 5}$ is

(A) 12

(B) 60

(C) 5

(D) 15

19. If $a = \sqrt{4 + \sqrt{15}} + \sqrt{4 - \sqrt{15}} - 2\sqrt{3 - \sqrt{5}}$ and $b = \sqrt{17 + 4\sqrt{13}} - \sqrt{17 - 4\sqrt{13}}$, then $\log_a b$ is

(A) 3

(B) 4

(C) 2

(D) 8

20. The value of $(\log_6 2)^3 + \log_6 8 \cdot \log_6 3 + (\log_6 3)^3$ is

(A) 1

(B) 2

(C) 0

(D) 3

21. If $a = \log_{245} 175$ and $b = \log_{1715} 875$, then $\frac{1 - ab}{a - b}$ is

(A) 7

(B) 3

(C) 1

(D) 5

22. If $(\log_b a)^3 + (\log_c b)^3 + (\log_a c)^3 = 3$, where a, b, c are real numbers greater than 1, then $\frac{ab + bc + ca}{a^2 + b^2 + c^2}$ equals

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{1}{3}$

(D) 3

23. If $\log_{2n}(1944) = \log_n(486\sqrt{2})$, then the number of significant digits in n^6 is (use $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$)

(A) 10

(B) 12

(C) 11

(D) 13

24. The number of values of x satisfying $x^{\log_7 13} + 13^{\log_{11} 17 \cdot \log_7 x} = 2 \cdot 17^{\log_{11} (13^{\log_7 x})}$ is

(A) 0

(B) 3

(C) 1

(D) 2

25. Let $x = (\log_{1/3} 5) (\log_{125} 343) (\log_{49} 729)$ and $y = 25^{3 \log_{289} 11 \cdot \log_{28} \sqrt{17} \cdot \log_{1331} 784}$. Then $x^2 + y^2$ is

(A) 41

(B) 50

(C) 25

(D) 34